FOR EDITH · OBSTETRICS & GYNAECOLOGY JUNIOR CLERKSHIP

Abortion & Post-Abortion Care

Tutorial | Week 1 · Foundations, ANC & early pregnancy | Gynaecology | Dr Nassimu

Abortion is the loss or termination of a pregnancy before fetal viability (28 weeks in Ugandan practice). Classify it by bleeding, pain, the cervical os, uterine size and tissue passed. Resuscitate first, then evacuate the uterus with MVA or misoprostol, treat sepsis aggressively, and never let a woman leave without post-abortion contraception.

Must know before the ward round

1. Classify by the os and the uterus: os closed + size = dates means threatened; os open means inevitable or incomplete; os closed + small uterus + bleeding settled means complete; os closed + fetus dead and retained means missed.
2. Stability first: shock index (pulse ÷ systolic BP) of 0.9 or more is abnormal and 1.4 or more is critical. Two large-bore cannulae, 1 L crystalloid, then reassess.
3. The speculum is the key examination: look at the os, remove products from the os with sponge forceps, and look for tears, burns, tenaculum marks or foreign bodies.
4. Incomplete abortion with uterine size under 14 weeks: MVA, or misoprostol 600 µg orally once, or 400 µg sublingually once (WHO 2022). Sharp curettage (D&C) is obsolete.
5. Uterus 14 weeks or more: misoprostol 400 µg sublingual, vaginal or buccal every 3 hours until expulsion, then check for retained tissue.
6. Missed miscarriage on ultrasound: CRL 7 mm or more with no heartbeat, or mean sac diameter 25 mm or more with no embryo. Below these cut-offs, rescan in 7–14 days.
7. Missed abortion under 14 weeks (WHO 2022): mifepristone 200 mg orally, then misoprostol 800 µg 1–2 days later; misoprostol 800 µg alone if no mifepristone.
8. Septic abortion: resuscitate, give IV antibiotics at once (ampicillin 2 g IV 6-hourly + gentamicin 5 mg/kg IV daily + metronidazole 500 mg IV 8-hourly), then evacuate within hours. Retained infected tissue is the source, so evacuation is the source control.
9. Bradycardia and hypotension out of proportion to the blood loss mean cervical shock: remove the products from the os with sponge forceps.
10. Closed os + empty uterus + positive pregnancy test with no previously confirmed intrauterine pregnancy is a pregnancy of unknown location (possible ectopic), not a complete abortion.
11. After unsafe abortion give tetanus toxoid 0.5 mL IM (± tetanus immunoglobulin 250 IU); give anti-D within 72 h to Rh-negative women after surgical evacuation or loss at 12 weeks or more.
12. Fertility returns within 2 weeks of an abortion. Offer contraception before discharge; almost every method, including an IUD, can start the same day unless there is sepsis. PAC is legal in Uganda and must never be delayed.

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1. Definition

Abortion is the ending of a pregnancy before the fetus can survive outside the uterus. Ugandan clinical practice uses **before 28 weeks**; WHO uses **before 22 weeks or under 500 g** for international comparison.

Term	Meaning
Spontaneous abortion (miscarriage)	Pregnancy loss with no deliberate intervention
Induced abortion	Pregnancy deliberately ended, by medication or by a procedure
Unsafe abortion (WHO)	A procedure to end a pregnancy done by someone without the necessary skills, or in an environment that does not meet minimal medical standards, or both. Examples: sticks, herbs, cassava stems, bicycle spokes, overdoses of chloroquine or other drugs, and "washing" with detergents
Recurrent miscarriage	Classically 3 or more consecutive losses (older RCOG definition, still used in much teaching). ESHRE and the 2023 RCOG update now use 2 or more
Early pregnancy loss / missed miscarriage (ultrasound terms)	**CRL 7 mm or more with no heartbeat**, or **mean sac diameter (MSD) 25 mm or more with no embryo**. Below these cut-offs, **rescan in 7–14 days** rather than acting
Pregnancy of unknown location (PUL)	Positive pregnancy test with no pregnancy seen in or outside the uterus. Includes complete miscarriage, very early intrauterine pregnancy (IUP) and ectopic pregnancy
Post-abortion care (PAC)	A package, not just a procedure: emergency treatment, counselling, **family planning before discharge**, other reproductive health services, and community links

The clinical types (threatened, inevitable, incomplete, complete, missed, septic) are in section 5.

> **UGANDA CONTEXT**
> Ugandan clinical teaching and the Uganda Clinical Guidelines define abortion as pregnancy loss **before 28 weeks**. Research and WHO statistics often use 22 weeks. A loss at 24 weeks is therefore still managed on the gynaecology (abortion) pathway in most Ugandan hospitals, while it would be a "late fetal loss" or "stillbirth" elsewhere.

2. Epidemiology

- **10–20% of clinically recognised pregnancies miscarry**, and about 80% of these losses happen before 12 weeks. The main cause is **aneuploidy** (an abnormal number of chromosomes), especially autosomal trisomy.
- Most spontaneous abortions happen in the first trimester and are chromosomal. Most **deaths**, however, come from **unsafe induced abortion**, through haemorrhage, sepsis and genital or bowel trauma.
- In Uganda, **abortion complications are a leading cause of gynaecological admission** and a consistent top cause of maternal death in MPDSR (maternal and perinatal death surveillance and response) reviews.
- **Unsafe abortion in Uganda:** national estimates (Prada et al., 2016) put induced abortions at roughly **300,000 a year**, with close to **100,000 women treated for complications** each year. Most of those women are young, poor, rural, unmarried students, or have an unmet need for contraception.

- **Determinants of septic abortion in Ugandan teaching hospitals** include rural residence, no income, induced abortion, evacuation attempted at home or at a lower-level unit, and no use of contraception.

Facility realities

- PAC with **misoprostol and MVA** is meant to be available at HC III and HC IV, and Ugandan trials show that **midwives achieve the same results as doctors** with misoprostol.
- In practice, MVA kits may be missing or unprocessed, misoprostol is out of stock, blood is short, and women arrive late because of fear of stigma, police or cost.
- An audit in a rural Ugandan hospital found **missing vital signs, poor fluid resuscitation of shocked women, and delayed evacuation of septic abortion**. These are exactly the gaps you can fix as a clerk.

> UGANDA CONTEXT
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> **Legal context (be precise and non-judgemental).** Article 22(2) of the Constitution says no one may terminate the life of an unborn child "except as may be authorised by law". The Penal Code criminalises procuring an abortion but protects a surgical operation done in good faith to preserve the mother's life. The MoH National Policy Guidelines and Service Standards for Sexual and Reproductive Health allow termination in specific situations, for example serious maternal illness, rape, incest or defilement. The 2015 MoH standards on reducing unsafe abortion were withdrawn soon after release, so the policy space is contested. **What is not contested: post-abortion care is legal, lifesaving and a patient right**. Treat every woman the same, whatever the cause, keep her information confidential, and never delay care to interrogate her.

3. Aetiology and Risk Factors

Causes of spontaneous loss

- **Chromosomal abnormality** (about half of first-trimester losses): trisomy (commonest is trisomy 16), then 45,X, then triploidy.
- **Maternal infection**: **malaria** (a febrile illness that inflames the placenta), **syphilis** (mainly later losses).
- **Uterine factors**: uterine anomalies, fibroids, **cervical insufficiency** (painless second-trimester loss).
- **Maternal disease**: uncontrolled diabetes or thyroid disease, antiphospholipid syndrome.

Risk factors for spontaneous loss

Non-modifiable	Modifiable
Maternal age over 35	Uncontrolled diabetes or thyroid disease
Previous miscarriage	Malaria, syphilis
Uterine anomalies	Smoking, alcohol
Antiphospholipid syndrome	Fibroids, cervical insufficiency (treatable in some)

Risk factors for unsafe induced abortion and its complications (Uganda)

- Young, poor, rural, unmarried, student, unmet need for contraception.
- Rural residence, no income, evacuation attempted at home or at a lower-level unit, no use of contraception (determinants of **septic** abortion).
- Late presentation because of fear of stigma, police or cost.

4. Pathophysiology

Why do most early pregnancies fail? About half of first-trimester miscarriages have a chromosomal abnormality. The embryo stops developing and hCG and progesterone fall. The **decidua (the pregnancy-altered endometrium) breaks down**, and prostaglandins are released, causing **cramping and cervical opening**. What you see clinically is the uterus trying to expel its contents.

Why does incomplete abortion bleed so much?

- After a pregnancy loss, bleeding normally stops through **myometrial contraction**. The "living ligatures" of the uterine muscle squeeze the spiral arteries shut.
- **Retained tissue keeps the uterus from contracting fully** and holds the placental bed open, so bleeding continues.
- The fix is therefore **mechanical**: empty the uterus with MVA, or chemically make it contract with misoprostol. Blood transfusion alone never solves the problem.

Cervical shock. Products stuck in the cervical canal stretch the cervix, which is richly supplied by parasympathetic (vagal) fibres. The result is a **vasovagal reflex**: bradycardia, hypotension and pallor, **out of proportion to the visible blood loss**. Removing the tissue from the os with sponge forceps reverses it within minutes.

Septic abortion

- Instruments or foreign bodies carry vaginal, bowel and soil organisms into a **necrotic, blood-filled uterus**, which is a perfect culture medium.
- Organisms are **polymicrobial**: *E. coli* and other Gram-negative rods, anaerobes (*Bacteroides*, *Peptostreptococcus*), streptococci (including group A), and ***Clostridium perfringens*** (gas gangrene, haemolysis, bronze or jaundiced skin, renal failure). *Clostridium tetani* also matters, so give **tetanus prophylaxis**.
- Infection spreads from **endometritis → myometritis → parametritis → pelvic peritonitis / abscess → septicaemia → septic shock and DIC** (disseminated intravascular coagulation).
- Antibiotics cannot penetrate dead tissue well, so **evacuation is source control**. This is why you evacuate promptly after starting antibiotics, rather than "cooling her off" for days.
- Unsafe procedures can **perforate the uterus and injure bowel or bladder**. Suspect this with peritonitis, free gas under the diaphragm, or bowel or fat seen in the vagina or aspirate.

Chromosomal error, infection, or unsafe interference | hCG

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B

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Decidual breakdown, prostaglandin release

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Cramps, cervical opening, expulsion

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Tissue retained: uterus cannot contract, placental bed open | Tissue stuck in the os: vagal reflex, cervical shock

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Ongoing haemorrhage, hypovolaemic shock | Instrumentation or foreign body seeds necrotic tissue

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Endometritis to parametritis to peritonitis

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Septicaemia, septic shock, DIC, AKI, death

Feature links

- Decidual breakdown and prostaglandins → crampy midline pain, cervical opening, vaginal bleeding.
- Open os with tissue still inside → inevitable or incomplete abortion.
- Retained tissue → poor myometrial contraction → heavy, persistent bleeding → tachycardia, hypotension, pallor, raised shock index.
- Tissue in the cervical canal → vagal stimulation → bradycardia and hypotension out of proportion to blood loss (cervical shock).
- Embryonic death without expulsion → closed os, uterus smaller than dates, no symptoms (missed abortion); prolonged retention → coagulopathy.
- Infected necrotic tissue → fever, rigors, offensive discharge, tender bulky uterus.
- Spread beyond the uterus → peritonism, cervical motion tenderness, pelvic abscess.
- *Clostridium perfringens* toxin → haemolysis → jaundice, bronze skin, dark urine, renal failure.
- Sepsis and tissue factor release → DIC → bleeding from drip sites, non-clotting blood.
- Instrument through the uterine wall → perforation → peritonitis, free gas, bowel or omentum in the vagina or aspirate.
- Retained tissue in a dirty wound → *C. tetani* → trismus and spasms days later.

How the drugs work (and why that explains their side-effects)

Drug	Mechanism	Consequences
Misoprostol	**Prostaglandin E1 analogue**: contracts the myometrium and softens (ripens) the cervix. Cheap, heat-stable, active by mouth	Side-effects are prostaglandin effects: **shivering, fever, diarrhoea, nausea and cramps**. **Sublingual** gives the fastest, highest peak; **vaginal** gives a longer effect with fewer side-effects
Mifepristone	**Progesterone receptor blocker**. Progesterone keeps the decidua alive and the uterus quiet; blocking it causes **decidual necrosis, cervical softening and a uterus much more sensitive to prostaglandins**	The combination (mifepristone first, then misoprostol 24–48 hours later) is more effective than misoprostol alone

Drug	Mechanism	Consequences
Oxytocin	Acts on myometrial oxytocin receptors	Works poorly in early pregnancy because the uterus has **few oxytocin receptors before about 20 weeks**. Prostaglandins, not oxytocin, are the drugs of choice for early losses

5. Classification / Staging

Type	Bleeding	Pain	Cervical os	Uterus vs dates	Tissue passed	Ultrasound
Threatened	Light to moderate	Mild or none	**Closed**	Equal to dates	None	Viable intrauterine pregnancy (IUP)
Inevitable	Heavy	Cramping, often strong	**Open**	Equal to or less than dates	None yet (membranes may have ruptured)	Sac low in the uterus or in the cervix
Incomplete	Heavy, ongoing	Cramping	**Open**; products may sit in the os	**Less than dates**	Partial	Retained products (heterogeneous material)
Complete	Settling or minimal	Settling	**Closed**	Small, firm	All (sac seen)	Empty uterus, thin endometrium
Missed (silent miscarriage)	Absent or brownish spotting	None	**Closed**	**Less than dates**	None	Embryo without heartbeat, or empty sac
Septic	Any, often offensive	Lower abdominal or pelvic pain	Usually open	Tender, bulky	Often incomplete	Retained products, with or without free fluid

Management is also staged by uterine size (WHO 2022): under 14 weeks versus 14 weeks or more (section 9).

6. Clinical Features

History

Take it while the drip is going in if she is bleeding heavily. Each question has a reason.

Ask	Why it matters (what changes the plan)
LMP, cycle regularity, gestational age; any earlier scan showing an intrauterine pregnancy (IUP)?	Dates the pregnancy. Uterine size smaller than dates suggests incomplete or missed abortion; size far larger than dates suggests a **molar pregnancy**, twins or fibroids. A previously confirmed IUP makes ectopic much less likely; no previous scan + closed os + empty uterus = PUL, not "complete abortion"
How much bleeding? (pads soaked per hour, clots, "flooding")	Soaking **1 pad or more in under an hour**, or passing large clots, means heavy bleeding. Look for shock; prepare for MVA and blood
Has any tissue passed? (what did it look like; did she bring it?)	Passed tissue suggests incomplete or complete abortion. Seeing a complete sac supports complete abortion; **inspect** anything she brought
Pain: crampy central, or one-sided and constant? Shoulder-tip pain? Fainting?	Crampy midline pain = the uterus is expelling. **Unilateral pain, shoulder-tip pain or collapse = ectopic pregnancy until proven otherwise**
Fever, rigors, offensive discharge; days since the event	Sepsis: antibiotics first, evacuation within hours. Late presentation (3 days or more) raises sepsis risk

Ask	Why it matters (what changes the plan)
"Did anything happen before the bleeding?" Asked gently and privately: any medicine, herb, insertion, or visit to anyone to "remove" the pregnancy?	Interference raises the risk of **sepsis, perforation, retained foreign body, tetanus** and drug toxicity: broaden cover, give tetanus prophylaxis, look for injury. Promise confidentiality, and explain it is to treat her safely, not to report her
Was this pregnancy planned? Current contraception? Plans for future pregnancy?	Opens PAC family-planning counselling and guides the method you offer. Unintended pregnancy is common after abortion
Past obstetric history, previous miscarriages, previous CS or uterine surgery	Recurrent loss work-up. A **previous CS** needs caution with repeated or high-dose misoprostol at larger uterine sizes (scar rupture), and raises perforation risk at MVA
Rhesus group if known; HIV status; sickle cell; bleeding disorder	Anti-D; PMTCT/ART linkage; anaemia risk and transfusion planning
Medical illness and allergies (diabetes, hypertension, epilepsy, asthma, bleeding disorder, penicillin allergy)	Choice of drug and anaesthesia: ergometrine is contraindicated in hypertension; antibiotic choice
Tetanus immunisation status	Td ± tetanus immunoglobulin after an unsafe procedure

Examination

Follow the order of Hutchison's Clinical Methods (24th edition), adapted for an emergency gynaecological examination. In a shocked woman, resuscitation (section 9) comes first and a quick bedside speculum to remove products from the os is part of resuscitation; complete the full sequence once she is stable.

Introduction, consent and preparation

- **Introduction and identity**: introduce yourself; confirm her name and age.
- **Explanation and consent**: explain each step, especially the speculum and bimanual examination; gain verbal consent. **Offer a female chaperone** and ensure privacy (screens, door closed).
- **Hand hygiene**: alcohol rub or soap and water; sterile gloves for the speculum if instrumentation is suspected.
- **Position and exposure**: supine with one pillow for the abdomen, exposing from the xiphisternum to the symphysis; then dorsal or lithotomy position for the speculum and bimanual. Lie her flat with legs raised if shocked.

General inspection and vital signs

Sign	Mechanism	What it means
Pale, sweating, confused, gasping, lying in a pool of blood	Hypovolaemia, hypoperfusion	Shock: shout for help and resuscitate now
Pallor	Blood loss, pre-existing anaemia	Check Hb; transfusion if under 7 g/dL or still shocked
Jaundice, bronze skin	Haemolysis (*Clostridium perfringens*)	Clostridial sepsis; check bilirubin, renal function
Dehydration	Vomiting, sepsis, poor intake	Fluids; watch gentamicin toxicity
Altered consciousness (AVPU)	Shock, sepsis, hypoglycaemia	Senior help; check glucose

Vital signs: pulse, BP, respiratory rate, temperature, SpO_2, capillary refill, level of consciousness (AVPU), urine output. Calculate the **shock index** (pulse ÷ systolic BP): **0.9 or more is abnormal, escalate; 1.4 or more is critical** (resuscitate and get blood). Young women keep their BP until late. Tachypnoea may mean sepsis, acidosis or ARDS.

Hands, face and neck

- **Hands**: cold, clammy peripheries and slow capillary refill (shock); palmar pallor (anaemia).
- **Face**: conjunctival pallor; scleral jaundice (haemolysis); bleeding gums (DIC).
- **Drip sites and skin**: oozing from cannula sites or bruising (DIC).
- **Neck**: brief check; neck stiffness or trismus days after an unsafe procedure suggests tetanus.

Chest and heart (brief)

Tachypnoea and crackles in sepsis or ARDS; flow murmur of anaemia.

Abdomen

Step	Sign	What it means
Inspection	Distension	Ileus, haemoperitoneum, bowel injury
Palpation	Uterus palpable above the symphysis (roughly 12 weeks or more); fundal height in weeks	Size vs dates
Palpation	**Guarding, rigidity, rebound tenderness**	Peritonitis: perforation, bowel injury, ruptured ectopic, pelvic abscess
Percussion	Shifting dullness	Haemoperitoneum (ruptured ectopic, perforation)
Auscultation	Absent bowel sounds	Ileus, peritonitis, bowel injury

Speculum (the single most useful step)

- Is the **os open or closed**? Width of the os.
- Are **products of conception (POC) in the os or vagina**? **Remove them now with sponge forceps** and keep them to inspect.
- Note the **source, amount and smell of the bleeding**, and any **offensive discharge** or pus.
- Look for **lacerations, burns or foreign bodies** (e.g. a stick, catheter or herbs) and **tenaculum marks on the cervix**, which are evidence of instrumentation.
- Other sources of bleeding: cervical polyp, ectropion, growth.

Bimanual

- **Uterine size in weeks** (this, not the LMP, decides MVA vs misoprostol), consistency, tenderness.
- **Cervical motion tenderness** (infection or ectopic).
- **Adnexal mass or tenderness** (ectopic, tubo-ovarian abscess).
- A **boggy or full pouch of Douglas** (blood, pus).

Completing the examination

- **Inspect any tissue** she passed or you removed (float it in saline: section 8).
- **Bedside tests**: urine hCG, Hb, bedside clotting test, mRDT, glucose (section 8).
- **Thank** her, cover her, **wash hands**.
- **Summarise** with the type of abortion, her stability, and whether there is sepsis or injury (model presentation in section 8).

Document the key negatives: "os closed/open, no POC in os, no offensive discharge, no cervical injury, uterus X weeks, non-tender, adnexae free, no peritonism".

Red flags

RED FLAG

- **Shock:** pulse over 110, systolic BP under 90, shock index 1 or more (1.4 or more critical), confusion, cold peripheries. Resuscitate before anything else.
- **Fever of 38 °C or more, rigors, foul discharge, uterine tenderness.** This is septic abortion: start antibiotics within the hour.
- **Peritonism, abdominal distension, absent bowel sounds, or bowel or omentum in the vagina or aspirate.** Suspect uterine perforation or bowel injury, which needs laparotomy.
- **Jaundice, dark urine, bronze skin, oliguria.** Suspect clostridial sepsis, haemolysis or acute kidney injury.
- **Bleeding from drip sites or gums, non-clotting blood.** Suspect DIC.
- **Closed os, empty uterus, positive pregnancy test and abdominal pain.** Think ectopic pregnancy; with signs of shock it is a ruptured ectopic until proven otherwise.
- **Uterus larger than dates with heavy bleeding, hyperemesis or grape-like vesicles.** Think molar pregnancy.
- **Bradycardia with hypotension and tissue in the os.** Cervical shock: remove the tissue now.
- **Fever with hypotension or confusion.** Septic shock: senior and anaesthetist now.

Call a senior at once for any of these.

Thinking like a clinician

- **Start with the os.**
  - **Closed os + uterus equals dates + viable pregnancy:** threatened.
  - **Open os:** the pregnancy cannot be saved (inevitable or incomplete).
  - **Closed os + small uterus:** either complete abortion, missed abortion or ectopic. **Scan before you conclude.**
- **"Complete abortion" is a diagnosis of evidence**, not of hope. Accept it when:
  - you saw a complete sac or products, **or**
  - a previous scan showed an IUP and the uterus is now empty, **or**
  - hCG falls appropriately.

  An **empty uterus without a previously confirmed IUP is a pregnancy of unknown location (PUL)** until proven otherwise.
- **Uterine size, not LMP, decides the method.** WHO bases incomplete-abortion treatment on **uterine size**: under 14 weeks vs 14 weeks or more. A woman at 16 weeks by dates whose uterus is 10-week size after passing most tissue can have MVA.
- **Every febrile woman with a recent pregnancy loss has septic abortion until proven otherwise.** Do not blame malaria alone. Do a malaria test, but still examine the pelvis.
- **Assume interference is possible in every case, without accusing anyone.** Tenaculum marks, cervical tears, foreign bodies, burns, or a woman who is very ill with an apparently "simple" miscarriage should change your plan: broader antibiotics, tetanus cover, a lower threshold for laparotomy.
- **If bradycardic and hypotensive, look in the os before giving more fluid.** Cervical shock reverses within minutes once the tissue is out.
- **Do not forget the second patient: her future.** Before she goes home, ask about her fertility intentions and whether she has a contraceptive method.
- **Common student mistakes:**

- Doing an MVA on a shocked patient before resuscitation.
- Missing an ectopic because the pregnancy test was "positive, so it must be an abortion".
- Giving misoprostol for a threatened abortion with a live fetus.
- Forgetting anti-D.
- Discharging without FP.
- Not examining the removed tissue.

7. Differential Diagnoses

Differential	Distinguishing clinical feature	Distinguishing investigation
Ectopic pregnancy	Unilateral pain (often before the bleeding), shoulder-tip pain, fainting, cervical excitation, adnexal tenderness or mass. The os is closed and the bleeding is scanty and dark	TVUS: empty uterus ± adnexal mass or free fluid; serial hCG
Molar pregnancy (gestational trophoblastic disease)	Uterus larger than dates, doughy; severe vomiting; early pre-eclampsia; passage of vesicles; theca-lutein cysts	Very high hCG; "snowstorm" appearance on ultrasound; histology of evacuated tissue
Cervical lesions (ectropion, polyp, cervicitis, cervical cancer)	Bleeding from a visible lesion; os closed; pregnancy normal	Speculum; scan confirms a viable IUP
Implantation bleeding	Light spotting around the expected period	Scan and dating
Septic abortion vs malaria / UTI / appendicitis	Uterine tenderness, offensive lochia and an open os point to the uterus	Speculum, pelvic exam, blood slide or mRDT, urinalysis, scan
Anovulatory or dysfunctional bleeding	Not pregnant	Negative urine or serum hCG
Uterine perforation with bowel injury (after an unsafe procedure)	Peritonitis, distension, free gas, faeculent aspirate	Erect chest X-ray / abdominal ultrasound; laparotomy

8. Clinical Workup

This is the order in which a senior on the Mulago or Mengo gynae emergency ward works up a woman with **bleeding in early pregnancy**. Resuscitation, bloods and removal of POC from the os happen within the first 15–30 minutes; she should not wait for a scan or a lab result before resuscitation or before POC are removed from the os.

Bedside

Test	Why / when	Expected finding	How it changes management	Available at
Urine pregnancy test (hCG)	Everyone	Positive confirms pregnancy. May stay positive for **2–4 weeks** after complete abortion	A positive test does not mean retained tissue; persisting positivity beyond 3–4 weeks with symptoms suggests retained tissue, ectopic or gestational trophoblastic disease	All levels
HemoCue / bedside Hb	Everyone	Baseline; **Hb under 7 g/dL**	Transfuse if under 7 g/dL or ongoing shock	HC III and above

Test	Why / when	Expected finding	How it changes management	Available at
Bedside clotting test	Sepsis, prolonged retention of a dead fetus, bleeding from sites	2 mL of blood in a plain glass tube held in the hand: **no clot by 7 minutes**, or a clot that breaks down, suggests **DIC**	Blood products, senior and anaesthetist; clotting screen if available	Any HC IV without a laboratory
mRDT for malaria	Fever plus pregnancy loss	Positive or negative	Treat malaria if positive, **but still examine the pelvis** for sepsis	All levels
Random blood sugar	Septic or confused women; after quinine	Hypoglycaemia	Dextrose	HC III and above (variable)
Urine dipstick	Everyone	Nitrites/leucocytes (UTI); blood; ketones (dehydration)	Treat UTI; fluids	All levels
Point-of-care ultrasound (if you are trained)	Unclear type, shock with closed os	Free fluid in the abdomen or pouch of Douglas (ruptured ectopic or perforation), retained products, fetal heartbeat	Laparotomy; evacuation; ectopic pathway	Where a trained user and machine exist
Inspect the tissue she passed or you removed	Every case with tissue	Float it in saline: villi look fluffy and fronded; a complete sac supports complete abortion; vesicles suggest mole	Confirms IUP; mole → hCG follow-up; no villi → think ectopic	All levels

CLINICAL PEARL

Bedside clotting time. Put 2 mL of venous blood in a plain glass tube and hold it in your hand. If it has **not clotted by 7 minutes** (or clots and then breaks down easily), suspect coagulopathy. This works at any HC IV without a laboratory.

Basic lab

Test	When	Expected / abnormal finding	How it changes management	Available at
Hb / full blood count	Everyone	Low Hb; **WBC raised** in sepsis; **platelets under 100** suggest DIC	Transfusion; supports sepsis; triggers a clotting screen	Hb at HC III+; FBC at HC IV and hospitals
Blood group, Rhesus D, crossmatch	Everyone	Rh-negative	Anti-D within 72 h after surgical evacuation or loss at 12 weeks or more; blood ready if heavy bleeding	HC IV and above can crossmatch
HIV and syphilis tests	Everyone	Positive	ART and PMTCT linkage; benzathine penicillin; syphilis explains second-trimester loss	All levels
Urea, creatinine, electrolytes	Sepsis, shock, oliguria, on gentamicin; repeat after 24 h in septic abortion	Rising creatinine = AKI	Adjust or stop gentamicin; fluid balance; dialysis referral	Some HC IVs and hospitals

Test	When	Expected / abnormal finding	How it changes management	Available at
Lactate	Septic abortion, where available	Raised lactate supports septic shock	HDU, fluids, vasopressors where available	Referral hospitals

In septic abortion, repeat Hb and creatinine after 24 h.

Imaging

Test	When	Expected / abnormal finding	How it changes management	Available at
Pelvic ultrasound (TVUS if available)	Unsure of the type, closed os, suspected missed abortion, ectopic, mole or perforation	Viable IUP (threatened); **CRL 7 mm or more with no heartbeat or MSD 25 mm or more with no embryo** (missed); heterogeneous retained products; **empty uterus** ± adnexal mass/free fluid; snowstorm (mole)	Confirms the type; an empty uterus without a prior IUP → **serial hCG** and ectopic pathway; below the cut-offs → rescan in 7–14 days	Most HC IVs (abdominal); all referral hospitals
Erect chest X-ray / abdominal X-ray	Peritonism after an unsafe procedure	**Gas under the diaphragm**, foreign body	Laparotomy	General hospitals and above

Specialised / referral-level

Test	When	Expected / abnormal finding	How it changes management	Available at
Serum β-hCG (quantitative)	PUL, suspected ectopic or mole	Suboptimal rise or plateau (ectopic); very high (mole); falling by at least 50% in 48 h (failing pregnancy)	Decides discharge vs ectopic management	Mainly regional/national referral and private labs
Blood culture, high vaginal swab	Septic abortion, before antibiotics if it causes no delay	Gram-negatives, anaerobes, *Clostridium*	Narrows antibiotics if she is not improving	Referral hospitals only in most of Uganda
Clotting screen (PT, APTT, fibrinogen)	Septic shock, bleeding from sites, abnormal bedside test	Prolonged PT/APTT, **fibrinogen under 2 g/L**	Fresh frozen plasma, platelets; senior and anaesthetist involvement	Regional/national referral
Bilirubin, LFTs	Jaundice, suspected clostridial sepsis or haemolysis	Raised unconjugated bilirubin	Supports haemolysis; consider early hysterectomy if the uterus is necrotic	Hospitals
Histology of tissue	Tissue looks abnormal (vesicles), no villi seen, or recurrent losses	Chorionic villi (confirms IUP); hydropic villi (mole)	Molar pregnancy → hCG follow-up; no villi → think ectopic	Referral hospitals

Putting it together

Model one-line summary

> "A 26-year-old P2+1 at 11 weeks by dates with 1 day of heavy bleeding and cramps, **haemodynamically stable** (pulse 100, BP 110/70, shock index 0.9, afebrile), Hb 9.5 g/dL, with an **open os, POC in the os** and a **9-week-size non-tender uterus**. **Working diagnosis: incomplete abortion, not septic, no evidence of**

> **interference.**"

Problem list

1. Incomplete abortion, uterine size under 14 weeks.
2. Mild-moderate anaemia (Hb 9.5 g/dL).
3. Rhesus status unknown.
4. Needs contraception (PAC).

Plan: POC removed from the os; MVA under paracervical block after ibuprofen (or misoprostol 600 µg orally if she prefers and can return); group and Rhesus; HIV/syphilis; iron; FP method before discharge.

Model case presentation (ward round)

> "Mrs N, a 24-year-old P1+1 at 10 weeks by dates, presented with 2 days of per vaginal bleeding and lower abdominal cramps, with passage of clots. She is **haemodynamically stable** (pulse 96, BP 112/70, afebrile), Hb 9.8 g/dL. On speculum the **os is open with products in the os**, and the uterus is 8-week size, non-tender. **Impression: incomplete abortion, not septic.** Plan: POC removed from the os; MVA under paracervical block with ibuprofen; group and Rhesus; HIV test; **post-abortion family planning counselling, she has chosen an implant**; review before discharge."

Worked example 1: using the speculum to classify

Speculum and bimanual findings	Classification	Next step
Os **closed**, small amount of blood from the os, uterus = dates, non-tender	**Threatened** (confirm a viable IUP on scan)	Reassure, paracetamol, return advice. No misoprostol, no MVA
Os **open**, membranes bulging or no tissue passed yet, uterus = dates	**Inevitable**	Evacuate (MVA or misoprostol by uterine size)
Os **open**, tissue in the os, uterus smaller than dates	**Incomplete**	Remove POC; MVA or misoprostol 600 µg oral
Os **closed**, bleeding settled, uterus small and firm, **she saw a sac pass or a previous scan showed an IUP**	**Complete**	No evacuation; PAC package
Os **closed**, bleeding settled, uterus empty on scan, **no prior IUP**	**PUL** (not "complete")	Serial hCG 48 h apart; ectopic safety-net
Os open, **offensive pus**, tender bulky uterus, fever	**Septic**	Antibiotics within 1 h, evacuate within hours
Os open, **tenaculum marks and a cervical tear**, peritonism	**Septic, induced, with possible perforation**	Antibiotics, resuscitation, erect CXR, laparotomy

Worked example 2: interpreting a scan report. A 30-year-old at 10 weeks by dates has spotting. Os closed. Scan: intrauterine sac, **MSD 18 mm, no embryo**.

- This is **below** the 25 mm cut-off, so you **cannot yet diagnose missed miscarriage** (dates may be wrong).
- Plan: **rescan in 7–14 days**. If she then has MSD 25 mm or more with no embryo, or an embryo with **CRL 7 mm or more and no heartbeat**, diagnose missed miscarriage and offer mifepristone + misoprostol, misoprostol alone, or MVA.
- Never give misoprostol "to be safe" on a single borderline scan: you might end a viable pregnancy.

9. Management

Immediate / emergency

First 5 minutes (any woman with bleeding in early pregnancy)

1. **Look at her from the door.** Is she pale, sweating, confused, gasping, lying in a pool of blood? If yes, **shout for help** (midwife, intern, anaesthetist) and go straight to resuscitation.
2. **Vitals now** and the **shock index** (0.9 or more = escalate; 1.4 or more = critical).
3. **Two large-bore cannulae (16–18 G)**; draw blood for **Hb, grouping & crossmatch, bedside clotting test, mRDT, HIV and syphilis** at the same time (plus cultures and renal function if septic).
4. **Quick speculum at the bedside** if bleeding is heavy or she is bradycardic: **remove POC from the os with sponge forceps**. This is both diagnostic and treatment (cervical shock).

ABCDE

- **A**: airway; call for help.
- **B**: **oxygen** if SpO_2 is low or she is shocked; respiratory rate (tachypnoea in sepsis, acidosis, ARDS).
- **C**: lie her flat, legs raised. **Warmed Ringer's lactate or normal saline, 1 L quickly**, then reassess pulse, BP and urine output. **Transfuse if Hb is under 7 g/dL, or if she is still shocked after 1–2 L of crystalloid.** Remove POC from the os. **Uterotonic** if bleeding is heavy and evacuation is not immediately possible: **misoprostol 400 µg oral/sublingual** or **ergometrine 0.2 mg IM** (not if hypertensive). Catheterise and start a fluid balance chart if shocked or septic.
- **D**: AVPU; glucose (septic, confused, after quinine).
- **E**: temperature; **fever 38 °C or more, rigors or foul discharge** → take cultures if possible but **give the first IV antibiotic doses within the hour**. Look for jaundice and bleeding from drip sites.

Then: decide the type (os, size, scan), look for **sepsis or injury**, evacuate by the method that fits the uterine size and her condition, give antibiotics if septic (IV) or after instrumentation, **anti-D, tetanus prophylaxis, analgesia**, and the **PAC package before discharge** (counselling, **contraception**, HIV/STI services, follow-up).

Definitive

Threatened abortion

1. Scan to confirm an **intrauterine pregnancy with a heartbeat**, and check for a subchorionic haematoma.
2. Reassure: most (about 80–90% with a live embryo) continue normally.
3. **Paracetamol** for pain. Avoid NSAIDs in ongoing pregnancy.
4. **No evidence supports strict bed rest** or routine hormones. Advise her to avoid heavy work and intercourse until the bleeding stops, mainly for comfort.
5. **Do not give misoprostol or do MVA.** The fetus is alive.
6. Treat any cause you find: malaria, UTI, syphilis.
7. Tell her to **return** if the bleeding becomes heavy, pain increases, tissue passes or she develops a fever. Continue ANC.

Inevitable and incomplete abortion: uterine size under 14 weeks

Offer a choice where she is stable and not septic. All three options are effective.

Option	Details	Success	Best for
MVA (vacuum aspiration)	Paracervical block + NSAID; done in the procedure room	~98–100%	Heavy bleeding, sepsis (after antibiotics), wants it done now, unreliable follow-up

Option	Details	Success	Best for
Misoprostol	**600 µg orally, single dose**, or **400 µg sublingually, single dose**	~90–96% within 1–2 weeks	Stable woman, no sepsis, can return; HC III with no MVA provider
Expectant	Wait for spontaneous expulsion	Lower and slower	Stable, minimal bleeding, strongly prefers to wait

- Review **7–14 days after misoprostol**. Base the decision on her **symptoms and examination**, not on the scan alone. A thin rim of endometrial tissue or clot does not need MVA if she is well.
- **Do MVA if** bleeding is heavy, she is febrile, or retained products persist and she has symptoms.

> EXAM FAVOURITE
> **"Why not D&C?"** WHO recommends that **sharp curettage be replaced by vacuum aspiration**. D&C causes more blood loss, more pain, more perforation and more intrauterine adhesions (Asherman syndrome), and it usually needs general anaesthesia and a theatre.

Incomplete abortion: uterine size 14 weeks or more

1. Resuscitate; start **IV antibiotics** if any sign of infection.
2. **Misoprostol 400 µg sublingual, vaginal or buccal every 3 hours** until the fetus and placenta are expelled. WHO sets no fixed maximum, but use clinical judgement and **lower doses or caution with a previous uterine scar**.
3. Alternative when misoprostol is unavailable or the cervix is already open: **oxytocin 20–40 IU in 1 L normal saline or Ringer's lactate**, titrated to contractions (receptors increase as pregnancy advances).
4. **After expulsion:** examine the placenta. If it is incomplete or bleeding continues, remove the retained tissue with sponge forceps or MVA (large cannula). **Dilatation and evacuation (D&E)** should be done only by trained providers.
5. **Oxytocin 10 IU IM** after expulsion helps the uterus contract.

> UGANDA CONTEXT
> Ugandan randomised trials at district hospitals showed that **misoprostol 400 µg sublingually every 3 hours (up to 5 doses)** for second-trimester incomplete abortion achieved complete evacuation within 24 hours in over 90% of women, **with midwives as safe and effective as doctors**. MoH policy lets midwives manage first-trimester PAC. Second-trimester care has traditionally been reserved for physicians, but task sharing is expanding.

Missed abortion (early pregnancy loss with closed os)

Regimen	Details
Preferred (WHO 2022), under 14 weeks	**Mifepristone 200 mg orally**, then **1–2 days later misoprostol 800 µg** vaginally, sublingually or buccally
Misoprostol alone (mifepristone not available)	**Misoprostol 800 µg** vaginally, sublingually or buccally; repeat doses may be given if needed (commonly 3-hourly, up to 3 doses)
Surgical	**MVA** under paracervical block (up to 12–14 weeks' uterine size, depending on the provider and cannula)
14–28 weeks (fetal death)	See the *Intrauterine fetal death* topic: mifepristone 200 mg, then misoprostol 400 µg every 4–6 hours

Complete abortion

1. Confirm it (section 6, *Thinking like a clinician*). If there was no previous scan showing an IUP, **scan, and consider serial hCG** to exclude ectopic.

2. No evacuation needed. Give anti-D if Rh-negative (below), analgesia, and the **PAC package**.

Septic abortion (a medical emergency)

1. **Resuscitate** (as above).
   - Aim for MAP of 65 mmHg or more and urine output over 0.5 mL/kg/h.
   - Catheterise and keep a **fluid balance chart**.
   - Take **blood cultures and a high vaginal swab** if you can, but **do not delay antibiotics**.
2. **IV antibiotics within 1 hour**: **ampicillin 2 g IV 6-hourly + gentamicin 5 mg/kg IV once daily + metronidazole 500 mg IV 8-hourly** (or the common alternative, ceftriaxone 1–2 g IV daily + metronidazole; drug table below).
   - **Continue IV until she has been fever-free for 48 hours**, then switch to oral **amoxicillin 500 mg 8-hourly** (or amoxicillin–clavulanate) **+ metronidazole 400 mg 8-hourly** to complete **7–14 days**.
   - Base the switch and the total duration on her response.
3. **Evacuate the uterus promptly**, within a few hours of starting antibiotics, **by MVA**.
   - Do not wait days for the fever to settle: **the retained tissue is the source**.
   - In large or second-trimester uteri, use misoprostol plus evacuation.
4. **Tetanus prophylaxis** after any unsafe procedure:
   - **Tetanus toxoid (TT/Td) 0.5 mL IM**.
   - Plus **tetanus immunoglobulin 250 IU IM** if she is unimmunised or her status is unknown and the wound is dirty.
5. **Look for injury and act:**
   - **Laparotomy** if there is peritonitis, suspected perforation or bowel injury, a pelvic abscess, or no improvement 24–48 hours after evacuation and antibiotics.
   - **Hysterectomy** may be lifesaving if the uterus is necrotic (e.g. clostridial infection) or there is uncontrollable haemorrhage.
6. **Supportive care:** analgesia, correct anaemia, watch for **DIC**, acute kidney injury and ARDS; involve anaesthesia / HDU.
7. **Refer** from HC IV to a regional or national referral hospital (e.g. Mulago, Kawempe) if she needs ICU, dialysis, or surgery beyond local capacity. **Start antibiotics and fluids before transfer.**

> **DRUG DOSING**
> **Gentamicin 5 mg/kg once daily** (e.g. 300 mg for a 60 kg woman). Reduce or stop it if urine output falls or creatinine rises, because it is nephrotoxic and ototoxic. In sepsis with acute kidney injury, ceftriaxone + metronidazole is a practical alternative.

Drug table

Drug	Dose (adult)	Route	Frequency	Duration	Key cautions
Misoprostol (incomplete, uterus under 14 weeks)	600 µg oral **or** 400 µg sublingual	Oral / SL	Single dose	Review at 7–14 days	Shivering, fever, diarrhoea, cramps; never for threatened abortion with a live fetus
Misoprostol (incomplete, uterus 14 weeks or more)	400 µg	SL, vaginal or buccal	Every 3 h	Until expulsion (Ugandan trials: up to 5 doses)	Lower doses or caution with a previous uterine scar

Drug	Dose (adult)	Route	Frequency	Duration	Key cautions
Misoprostol (missed abortion under 14 weeks)	800 µg (after mifepristone, or alone)	Vaginal, SL or buccal	Once; if alone, may repeat (commonly 3-hourly, up to 3 doses)	Until expulsion	As above
Misoprostol (uterotonic while awaiting evacuation)	400 µg	Oral / SL	Once	Once	Not a substitute for evacuation
Mifepristone	200 mg	Oral	Once, 1–2 days before misoprostol	Once	Missed abortion; often unavailable
Oxytocin (14 weeks or more)	20–40 IU in 1 L normal saline or Ringer's lactate	IV infusion	Titrated to contractions	Until expulsion	Poor effect before about 20 weeks (few receptors)
Oxytocin (after expulsion)	10 IU	IM	Once	Once	Helps the uterus contract
Ergometrine	0.2 mg	IM	Once	Once	**Avoid in hypertension**
Ampicillin	2 g	IV	6-hourly	Until fever-free 48 h	Covers streptococci, *Clostridium*, enterococci
Gentamicin	5 mg/kg (e.g. 300 mg for 60 kg)	IV	Once daily (24-hourly)	Until fever-free 48 h	Nephrotoxic, ototoxic; check renal function and hydration; reduce or stop if oliguric
Metronidazole	500 mg	IV	8-hourly	Until fever-free 48 h	Anaerobes
Ceftriaxone (common alternative)	1–2 g	IV	Once daily	Until fever-free 48 h	Widely used in Ugandan hospitals; **still add metronidazole**; practical if AKI
Amoxicillin (or amoxicillin–clavulanate)	500 mg	Oral	8-hourly	To complete 7–14 days	Step-down after 48 h fever-free
Metronidazole (oral)	400 mg	Oral	8-hourly	To complete 7–14 days	No alcohol
Tetanus toxoid (TT/Td)	0.5 mL	IM	Once	Once	After any unsafe procedure
Tetanus immunoglobulin	250 IU	IM	Once	Once	If unimmunised or status unknown and the wound is dirty
Anti-D immunoglobulin	250 IU before 20 weeks; 500 IU from 20 weeks (UK doses); Ugandan stock vial usually 1,500 IU	IM	Once, within 72 h	Once	Rh-negative after surgical evacuation or loss at 12 weeks or more; not required for medical abortion under 12 weeks (WHO 2022)

Drug	Dose (adult)	Route	Frequency	Duration	Key cautions
Ibuprofen	400–800 mg	Oral	30–60 min before MVA or with misoprostol	Short course	NSAIDs do not reduce misoprostol's effect; avoid in ongoing pregnancy (threatened abortion)
Paracetamol	1 g	Oral	6–8-hourly	As needed	Max 4 g/day
Lidocaine 1% (paracervical block)	Up to 20 mL (max **4.5 mg/kg** without adrenaline)	Cervical injection	Once before MVA	Once	See *MVA kit* topic
Diclofenac	75 mg	IM	Once (repeat as needed)	Short	Moderate to severe pain
Pethidine	50–100 mg	IM	As needed	Short	Severe pain; watch for respiratory depression

WHO recommends **against routine general anaesthesia** for vacuum aspiration. A paracervical block plus an NSAID, with calm verbal support ("vocal local"), is enough for most women.

Referral criteria

Level	What you can do	Refer up when
HC III	Diagnose; resuscitate (IV fluids); remove POC from the os; **misoprostol** for incomplete abortion under 14 weeks; MVA by a trained midwife or clinical officer; first-dose antibiotics; FP	Shocked, septic or second-trimester cases → HC IV or hospital
HC IV / General Hospital	All of the above plus **MVA**, second-trimester misoprostol, **blood transfusion**, **laparotomy** (perforation, ectopic), IV antibiotics	Needs ICU/HDU, dialysis, hysterectomy for a necrotic uterus, bowel surgery or surgery beyond local capacity → RRH
Regional Referral Hospital (RRH)	ICU / HDU, hysterectomy for necrotic uterus, bowel surgery, D&E, specialist ultrasound, management of complex complications	Dialysis or ICU unavailable, or complex complications beyond capacity → NRH
National Referral Hospital (Mulago, Kawempe)	Dialysis, ICU, all specialist surgery	n/a

Level split between RRH and NRH adapted from the source notes Verify *against UCG 2023 levels of care*.

Before any referral: IV line and fluids running, first antibiotic dose given, uterotonic if bleeding, POC removed from the os, a referral note with **vitals and drugs given**, and a health worker escorting her.

Monitoring and follow-up

- **In hospital**: pulse, BP, temperature, bleeding and urine output; fluid balance chart if shocked or septic; repeat Hb and creatinine after 24 h in sepsis.
- **Follow-up 7–14 days** after misoprostol, or 1–2 weeks after MVA.
- Check: bleeding, pain, fever; contraception satisfaction; results (HIV, syphilis); histology.
- **Pregnancy test** at 3–4 weeks only if symptoms persist. Persistent positive hCG suggests retained tissue, ectopic, or gestational trophoblastic disease.

Rhesus (anti-D) prophylaxis

- **Rh-negative women having surgical evacuation**, and any Rh-negative woman losing a pregnancy **at 12 weeks or more**: give **anti-D immunoglobulin 250 IU IM** before 20 weeks (500 IU from 20 weeks), **within 72 hours**.
- WHO 2022 says **anti-D is not required for medical abortion under 12 weeks**.

> **UGANDA CONTEXT**
> Anti-D is expensive and often unavailable in public units. Check the Rhesus group anyway, document it, and if anti-D cannot be given, **tell her** so the next pregnancy is monitored for isoimmunisation.

PAC elements: the complete package

The PAC consortium model has **5 elements**:

1. **Community and service-provider partnerships.** Prevent unwanted pregnancy and unsafe abortion; mobilise women to seek care early.
2. **Counselling.** Address her emotional and physical health needs; explain what happened, the warning signs, and when she will be fertile again.
3. **Treatment** of incomplete and unsafe abortion and its complications: MVA, misoprostol, antibiotics, surgery.
4. **Contraceptive and family-planning services. Ovulation can return within 8–10 days**, so start contraception **before she leaves**.
5. **Reproductive and other health services:** HIV testing and linkage, syphilis and STI treatment, cervical cancer screening (VIA), nutrition and iron, screening for gender-based violence, and referral for infertility or recurrent-loss work-up.

Post-abortion family planning (WHO MEC)

Method	After 1st-trimester abortion	After 2nd-trimester abortion	After septic abortion
COC, POP, injectable (DMPA), implant	**Start immediately** (MEC 1)	Immediately (MEC 1)	Immediately (MEC 1)
Copper IUD / LNG-IUS	**Immediately after MVA** (MEC 1)	MEC 2 (higher expulsion)	**Do not insert** (MEC 4). Wait until the infection has resolved
Condoms	Immediately; dual protection	Immediately	Immediately
Tubal ligation	Can be done at the time, if she chose it freely and was counselled before the emergency	Same	**Delay** until the infection has resolved
Fertility-awareness methods	Delay until regular cycles return	Delay	Delay

> **CLINICAL PEARL**
> If she gets misoprostol, an **implant or injectable can be given the same day**. An IUD can go in once the abortion is confirmed complete at follow-up. A Ugandan study found that offering an IUD after medical management of incomplete abortion is acceptable to many women. Do not make her come back twice.

10. Complications

Disease-related

Complication	Clues	Action
Haemorrhage / hypovolaemic shock	Tachycardia, hypotension, ongoing bleeding	Resuscitate, remove POC, uterotonic, evacuate, transfuse
Cervical shock	Bradycardia and hypotension out of proportion to blood loss, tissue in the os	Remove POC with sponge forceps; lie flat; fluids
Retained products	Persistent bleeding, open os, fever	Repeat MVA or misoprostol
Sepsis / septic shock / DIC	Fever, tachycardia, hypotension, bleeding from sites	Antibiotics, evacuation, supportive care, blood products
Tetanus	Trismus, spasms, days after an unsafe procedure	Prevent with TT ± TIG; ICU care
Acute kidney injury	Oliguria, rising creatinine (sepsis, haemolysis, gentamicin)	Fluids, stop nephrotoxins, dialysis referral
Long-term	Asherman syndrome (adhesions), secondary infertility (tubal damage), chronic pelvic pain, ectopic in future, isoimmunisation, psychological distress	Counselling, FP, fertility referral

Treatment- or procedure-related

Complication	Clues	Action
Uterine perforation (MVA, D&E, or unsafe procedure)	Instrument passes further than expected; peritonitis	Observe if small and she is stable; laparotomy if bowel injury or bleeding
Bowel or bladder injury	Faeculent aspirate, peritonitis, haematuria	Laparotomy, surgical team
Asherman syndrome	Amenorrhoea or infertility later; more after sharp curettage	Use vacuum aspiration, not D&C
Misoprostol side-effects	Shivering, fever, diarrhoea, nausea, cramps	Reassure; paracetamol; fever after misoprostol is short-lived, but persistent fever needs a sepsis review
Scar rupture	Previous CS, repeated misoprostol at larger uterine sizes	Lower doses or caution
Gentamicin toxicity	Oliguria, rising creatinine, hearing loss	Reduce or stop; switch to ceftriaxone + metronidazole
Local anaesthetic toxicity	Perioral tingling, seizures, arrhythmia	Respect the 4.5 mg/kg lidocaine maximum

ll. Prognosis

- **Threatened abortion with a live embryo**: about 80–90% continue normally.
- **Incomplete abortion**: MVA succeeds in ~98–100%; misoprostol in ~90–96% within 1–2 weeks; expectant management is lower and slower.
- **Fertility** returns quickly: **ovulation can return within 8–10 days**, and fertility within 2 weeks.
- Deaths are mostly from **unsafe induced abortion** (haemorrhage, sepsis, trauma) and are largely preventable by early presentation, prompt resuscitation, antibiotics and evacuation.
- Long-term risks after septic or unsafe abortion: Asherman syndrome, tubal infertility, chronic pelvic pain, future ectopic pregnancy, isoimmunisation, psychological distress.

12. Prevention

- **Primary**: prevent unintended pregnancy (contraception, community partnerships), and treat malaria, syphilis, diabetes and thyroid disease before and in pregnancy.
- **Secondary**: early presentation and prompt PAC; misoprostol and MVA available at HC III/IV; task sharing to midwives.
- **Tertiary (prevent repeat unsafe abortion and complications)**: **contraception before discharge**, anti-D for Rh-negative women, tetanus prophylaxis, HIV/STI services, and referral for recurrent-loss work-up.
- **System fixes** (from the Ugandan audit): record vital signs, resuscitate shocked women properly, and do not delay evacuation of septic abortion.

13. Special Populations

- **Previous caesarean section or uterine surgery**: caution with repeated or high-dose misoprostol at larger uterine sizes (scar rupture); perforation risk at MVA.
- **Rhesus-negative women**: anti-D within 72 h after surgical evacuation or loss at 12 weeks or more; if anti-D is unavailable, tell her so the next pregnancy is monitored.
- **Hypertension**: avoid ergometrine; use misoprostol or oxytocin.
- **Septic abortion and contraception**: no IUD (MEC 4) and delay tubal ligation until the infection has resolved; offer an implant, injectable or pills.
- **HIV-positive women**: ART and PMTCT linkage; all contraceptive methods remain options.
- **Acute kidney injury in sepsis**: avoid or adjust gentamicin; ceftriaxone + metronidazole.
- **Second-trimester loss (14–28 weeks)**: misoprostol regimens differ (section 9); see also the *Intrauterine fetal death* topic.

14. Key Points

High-yield points

- Is she stable? Then what type? Then is there sepsis or injury?
- Classify by the **os** and the **uterus**; the speculum is the key examination.
- Shock index 0.9 or more abnormal, 1.4 or more critical.
- Bradycardia and hypotension with tissue in the os = **cervical shock**: remove it.
- **Uterine size**, not the LMP, decides MVA vs misoprostol (under 14 vs 14 weeks or more).
- Closed os + empty uterus + no prior IUP = **PUL**, not complete abortion.
- Septic abortion: antibiotics within 1 h, **evacuate within hours**.
- MVA replaces D&C.
- Anti-D, tetanus prophylaxis, and **contraception before discharge**.
- PAC is legal, lifesaving and a patient right.

Numbers & doses

Item	Value
Abortion definition (Uganda / WHO)	Before **28 weeks** / before 22 weeks or under 500 g
Shock index	**0.9 or more** abnormal; **1.4 or more** critical
Transfusion trigger	Hb **under 7 g/dL**, or still shocked after 1–2 L crystalloid

Item	Value
Missed miscarriage on scan	**CRL 7 mm or more** with no heartbeat, or **MSD 25 mm or more** with no embryo
Incomplete abortion, uterus under 14 weeks	MVA, or misoprostol **600 µg oral once** or **400 µg SL once**
Incomplete abortion, uterus 14 weeks or more	Misoprostol **400 µg SL/vaginal/buccal every 3 h** until expulsion
Missed abortion under 14 weeks	**Mifepristone 200 mg** orally, then **misoprostol 800 µg** 1–2 days later (or misoprostol 800 µg alone)
Uterotonic while awaiting evacuation	Misoprostol **400 µg** oral/SL or ergometrine **0.2 mg IM**
Septic abortion antibiotics	**Ampicillin 2 g IV 6-hourly + gentamicin 5 mg/kg IV daily + metronidazole 500 mg IV 8-hourly**
IV antibiotics until	Fever-free **48 h**, then oral to complete 7–14 days
Paracervical block	Lidocaine 1% up to **20 mL**; max **4.5 mg/kg** (plain)
Tetanus after unsafe abortion	**Td 0.5 mL IM ± TIG 250 IU IM**
Anti-D (Rh-negative)	Within **72 h**; UK 250 IU before 20 weeks; the Ugandan vial usually stocked is 1,500 IU
Bedside clotting test	No clot by **7 minutes** = coagulopathy
Follow-up after misoprostol	**7–14 days**
Return of ovulation	As early as **8–10 days**

Exam traps

- **"Positive hCG + bleeding = miscarriage."** Wrong: closed os + empty uterus with no previous IUP is a **PUL, possibly ectopic**. Do serial hCG.
- **"Bradycardia and hypotension, so give more fluids and wait for theatre."** Wrong: this is **cervical shock**. Remove POC from the os with sponge forceps now.
- **"Septic abortion: give antibiotics for 48 h to cool her off, then evacuate."** Wrong: start antibiotics, then **evacuate within hours**; the retained tissue is the source.
- **"Use the LMP to decide MVA vs misoprostol."** Wrong: **uterine size** on bimanual decides (under 14 vs 14 weeks or more).
- **"Fever after an abortion is malaria."** Wrong until you have **examined the pelvis**: septic abortion first.
- **"Oxytocin is the best drug for a first-trimester incomplete abortion."** Wrong: few oxytocin receptors before about 20 weeks; use **misoprostol**.
- **"Insert an IUD at the time of MVA for septic abortion."** Wrong: **MEC 4**. Offer an implant, injectable or pills instead.
- **"A small retained rim on the day-10 scan needs MVA."** Wrong if she is well: manage by **symptoms and examination**, not the scan alone.
- **"D&C is the standard evacuation."** Wrong: vacuum aspiration has replaced sharp curettage.
- **"Give misoprostol to be safe on a borderline scan."** Wrong: below the cut-offs, rescan in 7–14 days.

Questions seniors ask

Viva questions

1. *What decides MVA vs misoprostol?* Uterine size (under 14 weeks either works), her stability (heavy bleeding or sepsis favours MVA), her preference, and whether she can come back.

2. *What is cervical shock and how do you treat it?* A vagal response to POC stretching the cervix. Bradycardia and hypotension out of proportion to blood loss. Remove the POC with sponge forceps; lie her flat; give fluids.
3. *Name the 5 elements of PAC.* Community partnerships, counselling, treatment, contraception/FP, other reproductive health services.
4. *Why start IV antibiotics before evacuating a septic abortion, but evacuate soon?* Antibiotics reduce bacteraemia during instrumentation. Evacuation removes the infected necrotic tissue that antibiotics cannot sterilise.
5. *When can she start an IUD after abortion?* Immediately after first-trimester MVA (MEC 1), unless there is septic abortion (MEC 4).
6. *What is the WHO misoprostol dose for incomplete abortion under 14 weeks?* 600 µg orally or 400 µg sublingually, single dose.
7. *A woman bleeds in early pregnancy, the os is closed and the uterus is empty on scan. Diagnosis?* Pregnancy of unknown location. It could be a complete miscarriage, a very early IUP or an ectopic. Do serial hCG and do not discharge without a safety net.

On the ward round

- **What is the first thing you do for a woman bleeding heavily at 10 weeks?** Assess ABC and vitals, calculate the shock index, two large-bore cannulae with bloods, fluids, and a speculum to remove POC from the os.
- **How do you tell inevitable from incomplete abortion?** Both have an open os; in incomplete, **some tissue has already passed** and the uterus is usually smaller than dates.
- **What makes you suspect an induced (unsafe) abortion?** Tenaculum marks, cervical tears, burns, foreign bodies, a very sick woman after an apparently "simple" miscarriage, or peritonism.
- **When would you do a laparotomy in septic abortion?** Peritonitis, suspected perforation or bowel injury, pelvic abscess, gas under the diaphragm, or no improvement 24–48 h after antibiotics and evacuation.
- **What are the scan criteria for missed miscarriage?** CRL 7 mm or more with no heartbeat, or MSD 25 mm or more with no embryo; below that, rescan in 7–14 days.
- **What must she leave with?** A contraceptive method (if she wants one), HIV/syphilis results or linkage, anti-D if Rh-negative and indicated, danger-sign advice and a follow-up date.
- **When is an empty uterus reassuring?** Only if a previous scan showed an IUP, or she passed a sac you saw, or hCG falls appropriately.

Mnemonics

"TIC-MS": the types in the order the os and the uterus decide them

- **T**hreatened: **T**ight (closed) os, **T**rue-to-dates uterus, heartbeat present
- **I**nevitable / **I**ncomplete: os **I**s open
- **C**omplete: **C**losed, **C**ontracted small uterus, **C**alm (bleeding stopped)
- **M**issed: **M**ute (no symptoms), **M**acerated or dead embryo, closed os
- **S**eptic: **S**mell, **S**hivers, **S**ore uterus

The septic abortion antibiotic trio: "A-G-M" (like "a German"): **A**mpicillin 2 g 6-hourly + **G**entamicin 5 mg/kg daily + **M**etronidazole 500 mg 8-hourly, until 48 hours fever-free.

Misoprostol "6-4 under 14; 4 every 3 over 14"

- Under 14 weeks: **600** µg oral or **400** µg sublingual, once.
- 14 weeks or more: **400** µg every **3** hours.

PAC = "C-C-T-F-R": **C**ommunity, **C**ounselling, **T**reatment, **F**amily planning, **R**eproductive health services.

OSCE tips

EXAM FAVOURITE

PAC counselling station. Cover:

- what happened, and that it was not her fault if spontaneous;
- danger signs: fever, heavy bleeding, foul discharge, severe pain;
- fertility returns within 2 weeks;
- her contraceptive choice (offer all methods with pros and cons);
- HIV testing;
- follow-up date.

Keep your tone warm and non-judgemental; examiners mark it.

- **History station**: ask about interference privately and gently, explain confidentiality, and never accuse.
- **Examination station**: state that you would offer a chaperone, calculate the shock index aloud, and describe the speculum findings (os, POC, smell, injury) before the bimanual.
- **Presenting**: give the type, her stability, and whether there is sepsis or injury in one line, then the plan including FP.

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SELF-TEST

Flashcards

26 cards. Cover the right-hand column and test yourself.

#	Question	Answer
1	Define abortion as used in Ugandan clinical practice.	Loss or termination of pregnancy **before fetal viability**, taken as **before 28 weeks** in Uganda (WHO uses <22 weeks or <500 g for international comparison).
2	Threatened vs inevitable abortion: the single distinguishing sign?	The **cervical os**: **closed** in threatened (pregnancy may continue), **open** in inevitable (pregnancy cannot be saved).
3	Features of incomplete abortion	Heavy bleeding, cramping, **open os**, products may be seen in the os, **uterus smaller than dates**, some tissue already passed.
4	Features of missed abortion	Dead embryo/fetus retained; loss of pregnancy symptoms, brownish spotting or none, **closed os**, **uterus smaller than dates**; scan: CRL ≥7 mm without heartbeat or MSD ≥25 mm without embryo.
5	What is cervical shock and its treatment?	Vagal reflex from products of conception distending the cervical canal → **bradycardia + hypotension out of proportion to blood loss**. Treat by **removing the POC from the os with sponge forceps**, lie flat, IV fluids.
6	WHO 2022 misoprostol regimen for incomplete abortion, uterine size <14 weeks	**600 µg orally** single dose **or 400 µg sublingually** single dose. (MVA is the alternative.)
7	WHO misoprostol regimen for incomplete abortion ≥14 weeks' uterine size	**400 µg sublingual, vaginal or buccal every 3 hours** until expulsion (caution/lower doses with a uterine scar).
8	WHO preferred regimen for missed abortion <14 weeks	**Mifepristone 200 mg orally**, then 1–2 days later **misoprostol 800 µg** vaginal/sublingual/buccal. Misoprostol alone 800 µg is the alternative.
9	Septic abortion: first-line IV antibiotic trio (WHO/Uganda)	**Ampicillin 2 g IV 6-hourly + gentamicin 5 mg/kg IV 24-hourly + metronidazole 500 mg IV 8-hourly** until fever-free 48 h, then oral to complete 7–14 days.
10	Why must a septic abortion be evacuated promptly rather than after days of antibiotics?	Retained necrotic infected tissue is the **source**; antibiotics cannot sterilise it. Evacuation = **source control**. Start antibiotics first, then evacuate within hours.
11	Organism causing haemolysis, bronze skin, gas in tissues and renal failure after unsafe abortion	***Clostridium perfringens*** (gas gangrene). May need hysterectomy.
12	Two extra prophylactic measures after an unsafe (instrumented) abortion	**Tetanus toxoid 0.5 mL IM** (± tetanus immunoglobulin 250 IU if unimmunised) and **anti-D** if Rh-negative (where available).
13	Name the 5 elements of post-abortion care (PAC)	1. Community & service-provider partnerships 2) Counselling 3) Treatment of incomplete/unsafe abortion & complications 4) Contraceptive/FP services 5) Reproductive & other health services (HIV, STI, cervical screening, GBV).
14	How soon can ovulation return after an abortion?	As early as **8–10 days** – so contraception should be started **before discharge**.
15	Can an IUD be inserted immediately after MVA for incomplete abortion?	**Yes** after first-trimester abortion (MEC 1); MEC 2 after second trimester; **NOT after septic abortion (MEC 4)** – delay until infection resolved.

#	Question	Answer
16	Why is D&C (sharp curettage) no longer recommended?	WHO: replace with vacuum aspiration. D&C has more **blood loss, pain, perforation, Asherman syndrome**, usually needs GA/theatre. MVA is safer and can be done by midwives.
17	Maximum dose of lidocaine for paracervical block (plain)	**4.5 mg/kg** (about 200 mg = **20 mL of 1%** for most adults). Aspirate before injecting.
18	Bleeding in early pregnancy, closed os, EMPTY uterus on scan, no previous IUP seen – diagnosis?	**Pregnancy of unknown location** – could be complete miscarriage, very early IUP or **ectopic**. Serial hCG ± repeat scan; safety-net advice.
19	Why is oxytocin a poor drug for first-trimester uterine evacuation?	The uterus has **few oxytocin receptors before ~20 weeks**; prostaglandins (misoprostol) act regardless of gestation.
20	Is PAC legal in Uganda?	**Yes.** Uganda's abortion law is restrictive and contested, but **post-abortion care is legal, lifesaving and a patient right** – never withhold or delay it, keep confidentiality.
21	Uterotonic options for heavy bleeding in incomplete abortion while awaiting evacuation	**Ergometrine 0.2 mg IM** (avoid in hypertension) or **misoprostol 400 µg oral/SL**; remove POC from os; resuscitate.
22	Shock index: how is it calculated and what are the obstetric cut-offs?	**Pulse ÷ systolic BP. 0.9 or more** = abnormal, escalate; **1.4 or more** = critical, resuscitate and get blood. BP falls late in young women.
23	Ultrasound criteria for diagnosing missed miscarriage	**CRL 7 mm or more with no heartbeat**, or **mean sac diameter 25 mm or more with no embryo**. Below these, **rescan in 7–14 days** before acting.
24	Speculum signs that an abortion was instrumented (unsafe)	**Tenaculum marks**, cervical **tears**, **burns**, **foreign bodies** (stick, catheter, herbs), offensive discharge. Change the plan: broad antibiotics, tetanus prophylaxis, look for perforation.
25	Bedside clotting test: how to do it and what is abnormal?	2 mL venous blood in a **plain glass tube**, held in the hand. **No clot by 7 minutes** (or a clot that breaks down) = coagulopathy (think DIC in septic abortion).
26	Closed os, bleeding settled, empty uterus on scan, no previous scan. Is this a complete abortion?	**No: it is a pregnancy of unknown location.** Accept 'complete' only if a prior scan showed an IUP, a sac was seen passing, or hCG falls appropriately. Do **serial hCG 48 h apart** and give ectopic safety-net advice.

SELF-TEST

Single best answer questions

15 questions. The answers and explanations follow at the end.

1. A 23-year-old P0+1 at 9 weeks presents to a HC IV with light bleeding and mild cramps. Pulse 84, BP 118/74, afebrile. Speculum: closed os, minimal blood. Ultrasound: intrauterine gestational sac with an embryo, CRL 22 mm, heartbeat present. What is the MOST appropriate management?

- A Reassurance, paracetamol, return precautions and continue ANC
- B Manual vacuum aspiration
- C Misoprostol 400 µg sublingually
- D Admit for strict bed rest and progesterone injections
- E Ergometrine 0.2 mg IM

2. A 28-year-old P2+0 at 11 weeks by LMP has had heavy bleeding with clots for 6 hours. Pulse 48, BP 80/50, pale and sweaty. Estimated visible blood loss is moderate. On speculum, tissue is seen protruding through the os. What is the MOST appropriate immediate action?

- A Give atropine and wait for the pulse to recover
- B Give misoprostol 600 µg orally and observe
- C Transfer immediately to theatre for laparotomy
- D Remove the products from the os with sponge forceps while resuscitating
- E Perform an urgent ultrasound scan before anything else

3. A 19-year-old student presents 3 days after 'a procedure' at a private clinic for an unwanted pregnancy of 10 weeks. Temperature 39.2 °C, pulse 124, BP 88/56, offensive vaginal discharge, tender bulky uterus, open os. Abdomen is soft apart from suprapubic tenderness. After starting IV fluids, what is the BEST next sequence?

- A Oral amoxicillin and metronidazole, review in 1 week
- B IV ampicillin, gentamicin and metronidazole now, then MVA within hours once antibiotics are running
- C IV antibiotics for 48–72 hours until afebrile, then MVA
- D Immediate MVA before any antibiotics to remove the source
- E Laparotomy and hysterectomy

4. A 32-year-old P3+0 had MVA for septic incomplete abortion yesterday and is on ampicillin, gentamicin and metronidazole. Today she has worsening abdominal distension, absent bowel sounds, generalised rebound tenderness and the MVA aspirate is noted to have smelt faeculent. Erect CXR shows gas under the diaphragm. What is the MOST appropriate management?

- A Continue antibiotics and repeat MVA
- B Add IV ceftriaxone and observe 48 hours
- C Nasogastric tube and conservative management of ileus
- D Refer for CT scan before deciding
- E Resuscitate and perform laparotomy

5. A 25-year-old at 8 weeks' uterine size has an incomplete abortion. She is stable (pulse 88, BP 120/76, afebrile, Hb 11.2 g/dL), bleeding is moderate, and she lives 2 km away with a phone. The HC III has misoprostol but the only MVA-trained midwife is on leave. What is the MOST appropriate treatment?

A Refer to the regional hospital for D&C
B Misoprostol 400 µg sublingually every 3 hours until expulsion
C Misoprostol 600 µg orally as a single dose with review in 7–14 days
D Oxytocin 20 IU infusion
E Expectant management with no follow-up

6. A 30-year-old P2+1 at 17 weeks by dates presents with heavy bleeding; fetus was expelled at home 2 hours ago. Uterus is 16-week size, os open, placenta not delivered. Pulse 100, BP 110/70, afebrile. Which is the MOST appropriate uterotonic regimen to complete the evacuation?

A Misoprostol 600 µg orally single dose
B Misoprostol 25 µg vaginally 6-hourly
C Misoprostol 400 µg sublingually every 3 hours until the placenta is expelled, then check for retained tissue
D Mifepristone 200 mg and discharge
E Ergometrine 0.5 mg IV every 15 minutes

7. A 22-year-old presents with vaginal bleeding at 7 weeks by LMP. Urine hCG positive. The bleeding is scanty and dark, and she reports left iliac fossa pain that started 2 days before the bleeding. Speculum: closed os. Pulse 104. What is the MOST important diagnosis to exclude first?

A Ectopic pregnancy
B Complete abortion
C Threatened abortion
D Cervical ectropion
E Molar pregnancy

8. A woman had MVA for first-trimester incomplete abortion. She is afebrile, not septic, and wishes to avoid pregnancy for at least 3 years. Which contraceptive option is MOST appropriate to offer before she leaves the procedure room?

A Advise her to wait 6 weeks before starting any method
B Emergency contraception only
C Fertility-awareness method starting now
D Tubal ligation immediately without prior counselling
E Copper IUD inserted immediately after the MVA

9. A 26-year-old had septic incomplete abortion evacuated today. She wants an IUD. What is the MOST appropriate advice?

A Insert a copper IUD now to reduce future infection
B Insert an LNG-IUS now
C IUDs are contraindicated for life after septic abortion
D Defer IUD until infection has resolved; offer an implant or injectable now
E Only condoms are allowed for 6 months

10. A 34-year-old G4P3 at 12 weeks is found on scan to have an embryo with CRL 30 mm and no cardiac activity. She has only brown spotting and a closed os. She is stable and requests medical management. Mifepristone is available. Which regimen is BEST? Advanced

- A Misoprostol 600 µg orally once
- B Mifepristone 200 mg orally, then misoprostol 800 µg vaginally/sublingually/buccally 1–2 days later
- C Misoprostol 25 µg vaginally 4-hourly
- D Oxytocin infusion
- E Ergometrine 0.2 mg IM daily for 3 days

11. A 21-year-old presents with fever 38.8 °C 4 days after passing tissue at home at 9 weeks. The clinician finds a positive malaria RDT and plans to discharge her on artemether-lumefantrine. What is the MOST important omission?

- A Checking her HIV status
- B Speculum and bimanual examination for septic abortion
- C Giving iron supplements
- D Repeating the malaria test by microscopy
- E Checking her blood group

12. A 29-year-old P1+0 is Rhesus D-negative and has MVA for incomplete abortion at 11 weeks. Anti-D is available. What should you do? Advanced

- A No anti-D is ever needed before 12 weeks
- B Give anti-D only in the next pregnancy
- C Give anti-D 1500 IU IV only if Kleihauer is positive
- D Give anti-D 250 IU IM within 72 hours
- E Give anti-D only if the partner is Rh-positive and she consents to paternity testing

13. At an HC III, a 20-year-old with septic incomplete abortion at 10 weeks has pulse 130, BP 80/40, confusion and oliguria. The facility has no theatre, blood or IV gentamicin, but has IV fluids, ceftriaxone and metronidazole. What is the BEST action?

- A Start IV fluids, give IV ceftriaxone and metronidazole, remove POC from os, and refer urgently with an escort
- B Perform MVA at the HC III immediately and keep her there
- C Refer immediately without starting any treatment to save time
- D Give oral amoxicillin and observe overnight
- E Give misoprostol 600 µg and send home

14. Which finding on history/examination MOST strongly suggests that an incomplete abortion was induced with instrumentation?

- A Uterus smaller than dates
- B Passage of clots
- C Tenaculum marks and a laceration on the anterior lip of the cervix
- D Lower abdominal cramping
- E Positive urine hCG

15. A 24-year-old at 10 weeks' uterine size with incomplete abortion receives misoprostol 600 µg orally. At review on day 10 she feels well, bleeding is light, she is afebrile, and ultrasound shows a 12 mm heterogeneous endometrial echo. What is the MOST appropriate management? Advanced

- A Immediate MVA because of retained products
- B Repeat misoprostol 600 µg and review in 1 week
- C Order serum hCG daily for a week
- D Start IV antibiotics
- E Reassure – treatment successful; no further intervention; confirm contraception

Answers & explanations

1. A · Reassurance, paracetamol, return precautions and continue ANC

This is a **threatened abortion** (closed os, live IUP). Correct: **reassure, analgesia (paracetamol), safety-net advice**, continue ANC; most continue normally. **B/C** would end a live pregnancy – absolutely wrong. **D**: strict bed rest and routine progestogens have no proven benefit in threatened miscarriage and hospital bed rest carries VTE and cost burdens. **E**: ergometrine contracts the uterus and could provoke expulsion; it has no role with a live fetus.

2. D · Remove the products from the os with sponge forceps while resuscitating

Bradycardia with hypotension out of proportion to bleeding and tissue at the os = **cervical shock** (vagal). **Remove the POC with sponge forceps** – usually reverses it within minutes – together with IV fluids and lying flat. **A**: atropine may be used if bradycardia persists but does not treat the cause. **C**: no evidence of intraperitoneal bleeding. **B**: misoprostol takes hours; the os is already open with tissue sitting there. **E**: scanning delays a lifesaving 30-second manoeuvre.

3. B · IV ampicillin, gentamicin and metronidazole now, then MVA within hours once antibiotics are running

Septic abortion with septic shock: **IV broad-spectrum antibiotics immediately, then prompt evacuation (source control) within hours**. **A**: oral outpatient therapy is dangerous in septic shock. **C**: waiting days leaves infected necrotic tissue in place – a classic audit failure in Uganda. **D**: evacuating before antibiotics increases bacteraemia; antibiotics take minutes to start. **E**: laparotomy/hysterectomy is for peritonitis, perforation, necrotic uterus or failure to respond – not first-line here.

4. E · Resuscitate and perform laparotomy

Peritonitis + free gas + faeculent aspirate after an unsafe procedure = **uterine perforation with bowel injury → resuscitate and laparotomy** (with surgical team; may need bowel repair/stoma ± hysterectomy). **A/B/C** treat symptoms while bowel contents soil the peritoneum. **D**: CT is unnecessary and unavailable in most units; the diagnosis is clinical + X-ray, and delay is lethal.

5. C · Misoprostol 600 µg orally as a single dose with review in 7–14 days

Stable, non-septic incomplete abortion <14 weeks: **misoprostol 600 µg PO single dose** (or 400 µg SL) is WHO-recommended with ~90–96% success; review at 7–14 days. **A**: D&C is obsolete and referral is unnecessary. **B**: the 3-hourly repeat regimen is for ≥14 weeks' uterine size. **D**: oxytocin works poorly in the first trimester (few receptors). **E**: expectant management is an option but must include follow-up and safety-netting.

6. C · Misoprostol 400 µg sublingually every 3 hours until the placenta is expelled, then check for retained tissue

Uterine size ≥14 weeks: **misoprostol 400 µg SL/PV/buccal every 3 hours** until expulsion, then inspect placenta and remove remaining tissue. **A**: single-dose 600 µg is for <14 weeks. **B**: 25 µg is a third-trimester labour induction dose – far too low here. **D**: mifepristone alone is slow and discharge with a retained placenta and heavy bleeding is unsafe. **E**: wrong dose/route; ergometrine standard is 0.2 mg IM/IV and it can trap the placenta.

7. A · Ectopic pregnancy

Unilateral pain **preceding** scanty dark bleeding, closed os and tachycardia = **ectopic pregnancy until proven otherwise** (the fatal miss). **B/C** are possible but less dangerous and the history fits ectopic better. **D**: ectropion causes painless contact bleeding. **E**: molar pregnancy usually gives uterus larger than dates and hyperemesis, not unilateral pain.

8. E · Copper IUD inserted immediately after the MVA

After uncomplicated first-trimester abortion, **immediate IUD insertion is MEC category 1** and ideal for long-acting protection. **A**: ovulation may return within 8–10 days – waiting 6 weeks exposes her to repeat pregnancy. **C**: fertility-awareness methods are unreliable until cycles are regular. **D**: sterilisation requires prior informed, unpressured counselling – not decided in an emergency. **B**: EC is not ongoing contraception.

9. D · Defer IUD until infection has resolved; offer an implant or injectable now

IUD after **septic abortion is MEC 4** – defer until infection resolves; meanwhile **hormonal methods (implant, DMPA, COC) can start immediately (MEC 1)**. **A/B** risk worsening pelvic infection. **C**: the restriction is temporary. **E**: all hormonal methods are safe now.

10. B · Mifepristone 200 mg orally, then misoprostol 800 µg vaginally/sublingually/buccally 1–2 days later

Missed abortion <14 weeks: WHO recommends **mifepristone 200 mg then misoprostol 800 µg 1–2 days later** (more effective than misoprostol alone). **A**: 600 µg single dose is the incomplete-abortion regimen; missed abortion needs higher dose as the os is closed and tissue fully attached. **C**: third-trimester induction dose. **D/E**: ineffective in early pregnancy.

11. B · Speculum and bimanual examination for septic abortion

Any fever after pregnancy loss is **septic abortion until proven otherwise** – examine the pelvis (offensive discharge, open os, tender uterus). Malaria can co-exist. **A/C/E** are good PAC components but not the dangerous omission. **D**: repeating the malaria test does not address the likely uterine source.

12. D · Give anti-D 250 IU IM within 72 hours

Surgical evacuation in an Rh-negative woman: **anti-D 250 IU IM within 72 h** (before 20 weeks). WHO says anti-D is not needed for *medical* abortion <12 weeks, but instrumentation increases fetomaternal haemorrhage. **A** overgeneralises. **C**: Kleihauer is used after 20 weeks to size the dose. **B**: prophylaxis must be given now. **E**: impractical and not standard – treat all Rh-negative women.

13. A · Start IV fluids, give IV ceftriaxone and metronidazole, remove POC from os, and refer urgently with an escort

Septic shock with organ dysfunction needs a higher-level facility, but **stabilise and start antibiotics before transfer**: IV fluids, first-dose antibiotics, remove POC from the os, then urgent escorted referral with a note. **B**: she needs HDU care, blood and theatre backup. **C**: untreated transfer is a common preventable cause of death. **D/E** are dangerously inadequate.

14. C · Tenaculum marks and a laceration on the anterior lip of the cervix

Tenaculum marks/cervical lacerations (and foreign bodies, vaginal burns) indicate instrumentation – raising risk of sepsis, perforation and tetanus. **A/B/D/E** occur in spontaneous miscarriage too.

15. E · Reassure – treatment successful; no further intervention; confirm contraception

Success after misoprostol is judged **clinically**: she is well with light bleeding and a thin (<15 mm) echo is expected – blood/decidua, not necessarily retained products. **A/B** overtreat and expose her to unnecessary procedures. **D**: no signs of infection. **C**: unnecessary; hCG testing at 3–4 weeks only if symptoms persist.

SELF-TEST

Clinical cases

Work through each step before reading the model answer.

Case 1: Heavy bleeding at 10 weeks at a HC IV

A 27-year-old P2+0 is brought to a HC IV in Mukono at 8 pm with heavy per vaginal bleeding and clots for 5 hours at about 10 weeks by LMP. She has soaked 4 pads in the last hour and feels dizzy.

1. What are your immediate priorities?

> Call for help; **ABC**; lie flat, legs raised, oxygen; **2 large-bore IV cannulae**; draw blood for **Hb, grouping & crossmatch, malaria test, HIV/syphilis**; **1 L warmed Ringer's/NS** rapidly; vitals every 15 min; catheter for urine output.

2. What focused history do you take while resuscitating?

> LMP and early scan; amount of bleeding and **tissue passed**; pain character (crampy vs unilateral/shoulder-tip); fever/foul discharge; **any interference** (asked privately, non-judgementally); previous CS; contraception and pregnancy intention; Rh group; HIV status; tetanus immunisation.

3. Pulse 118, BP 92/60, afebrile, Hb (HemoCue) 8.1 g/dL. Speculum: open os with tissue in the canal; uterus 8-week size, non-tender, no adnexal masses. What is the diagnosis and the immediate step?

> **Incomplete abortion with haemorrhagic shock (compensating)**, no sepsis. **Remove POC from the os with sponge forceps** immediately; give **misoprostol 400 µg SL** or **ergometrine 0.2 mg IM** (if normotensive) to contract the uterus; continue fluids; crossmatch 1–2 units.

4. What definitive treatment and supporting care?

> **MVA** once resuscitation is under way: **ibuprofen 400–800 mg PO** (if able to swallow) and **paracervical block (lidocaine 1%, ≤20 mL)**; appropriate cannula; examine the tissue (villi confirm IUP). Transfuse if Hb <7 or still unstable. **Anti-D** if Rh-negative. Oral iron.

5. Before discharge, what does the complete PAC package include?

> **Counselling** (what happened, danger signs, fertility returns within 2 weeks); **contraception chosen and started** (e.g. implant or immediate IUD – MEC 1); **HIV test and linkage**, syphilis screen; iron; **follow-up in 1–2 weeks**; community link (VHT).

Take-away: Resuscitate and clear the os first; MVA is definitive; PAC is not complete until she leaves with contraception and follow-up.

Case 2: Fever after an unsafe abortion

An 18-year-old S.4 student is brought by her aunt to a regional referral hospital with fever, lower abdominal pain and foul-smelling discharge 4 days after she 'removed a pregnancy' of about 12 weeks with the help of a local herbalist.

1. What history is most important and how will you ask it?

> Alone with her, with privacy and assurance of confidentiality: what was done (instruments, sticks, herbs, drugs – e.g. chloroquine, pessaries), when, bleeding, tissue passed, pain (worsening/generalised?), vomiting, diarrhoea, urine output, jaundice. Tetanus vaccination. LMP, contraception, sexual history (consent – consider defilement; she may be a minor; safeguarding).

2. Temp 39.5 °C, pulse 132, BP 84/50, RR 28, cold extremities, jaundiced. Abdomen: distended, generalised guarding, absent bowel sounds. Speculum: offensive discharge, open os, a laceration of the cervix. What is your assessment?

> **Septic abortion with septic shock**, probable **uterine perforation with peritonitis** (possible bowel injury); jaundice suggests **haemolysis (clostridial?)** or hepatic dysfunction/sepsis. Differentials for the jaundice: severe malaria, drug toxicity.

3. What investigations?

> FBC, **grouping/crossmatch**, blood culture, HVS, **urea/creatinine/electrolytes**, **bilirubin/LFTs**, **clotting (or bedside clotting time)**, malaria test, HIV, urinalysis, **erect CXR (free gas)**, abdominal/pelvic ultrasound (retained POC, free fluid, collections).

4. Outline management step by step.

> 1. Resuscitate: oxygen, 2 IV lines, crystalloid boluses, catheter, fluid chart. 2) **IV ampicillin 2 g 6-hourly + gentamicin 5 mg/kg daily (adjust for renal function) + metronidazole 500 mg 8-hourly** (or ceftriaxone + metronidazole) immediately. 3) **Tetanus toxoid 0.5 mL IM + TIG 250 IU**. 4) Blood products for DIC. 5) **Laparotomy** (peritonitis + suspected perforation) with surgical team: repair perforation/bowel, peritoneal lavage, evacuate uterus; **hysterectomy** if uterus necrotic/gangrenous. 6) HDU/ICU postoperatively.

5. What complications should you anticipate post-operatively and what follow-up?

> Septic shock, DIC, AKI (sepsis, haemolysis, gentamicin), ARDS, wound sepsis, pelvic abscess, tetanus, death. Later: infertility, chronic pelvic pain. Follow-up: psychosocial support, **contraception** (implant/DMPA now; no IUD while septic), HIV/STI care, school re-entry support, safeguarding referral if a minor.

> **Take-away:** Unsafe abortion kills through sepsis and hidden visceral injury – antibiotics, source control (evacuation or laparotomy) and tetanus cover, delivered without judgement.

Case 3: Painless bleeding: is it really a miscarriage?

A 31-year-old G3P2 at 7 weeks by LMP presents to Mengo casualty with 3 days of dark spotting. She says she passed 'something' yesterday. Pregnancy test positive. She has mild left lower abdominal discomfort.

1. What are the possibilities and what examination findings would help?

> Complete, incomplete, threatened or missed miscarriage; **ectopic pregnancy**; molar pregnancy. Examine: vitals (tachycardia, postural drop), abdominal tenderness/guarding, speculum (os open/closed, tissue), bimanual (cervical excitation, adnexal mass, uterine size).

2. Vitals normal. Os closed, minimal old blood, uterus bulky, mild left adnexal tenderness, no mass felt. TVUS: empty uterus, endometrium 9 mm, no adnexal mass, no free fluid. No previous scan. Diagnosis?

> **Pregnancy of unknown location (PUL)** – NOT 'complete abortion', because an IUP was never documented and she has adnexal tenderness. She did not bring the tissue.

3. What is your next investigation plan?

> **Serum hCG now and at 48 hours**. Fall >50% suggests failing pregnancy (complete miscarriage); rise >63% suggests an IUP (rescan when above discriminatory level); a suboptimal rise/plateau or slow fall suggests ectopic. Group and Rh, Hb.

4. hCG 1,850 IU/L, and 48 h later 1,920 IU/L; she now has more left-sided pain. What does this mean and what will you do?

> A **plateauing hCG above the discriminatory zone with an empty uterus = likely ectopic pregnancy**. Admit, repeat TVUS by an experienced sonographer, prepare for laparoscopy/laparotomy or methotrexate if criteria met (see Ectopic topic). Do not evacuate the uterus blindly.

5. What should she be told if she had been discharged earlier?

> **Safety-net advice**: return immediately with worsening one-sided pain, shoulder-tip pain, dizziness/fainting or heavy bleeding; attend for repeat hCG. Document clearly.

Take-away: Never call it 'complete abortion' without evidence of an IUP or passed sac – an empty uterus is a PUL, and ectopic must be excluded.

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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.